

PHD PROGRAM

Applied Data Science & Artificial Intelligence



UNIVERSITÀ
DEGLI STUDI
DI TRIESTE

Explainable Artificial Intelligence



APPLIED DATA SCIENCE &
ARTIFICIAL INTELLIGENCE



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- **LIME: INTRODUCTION**

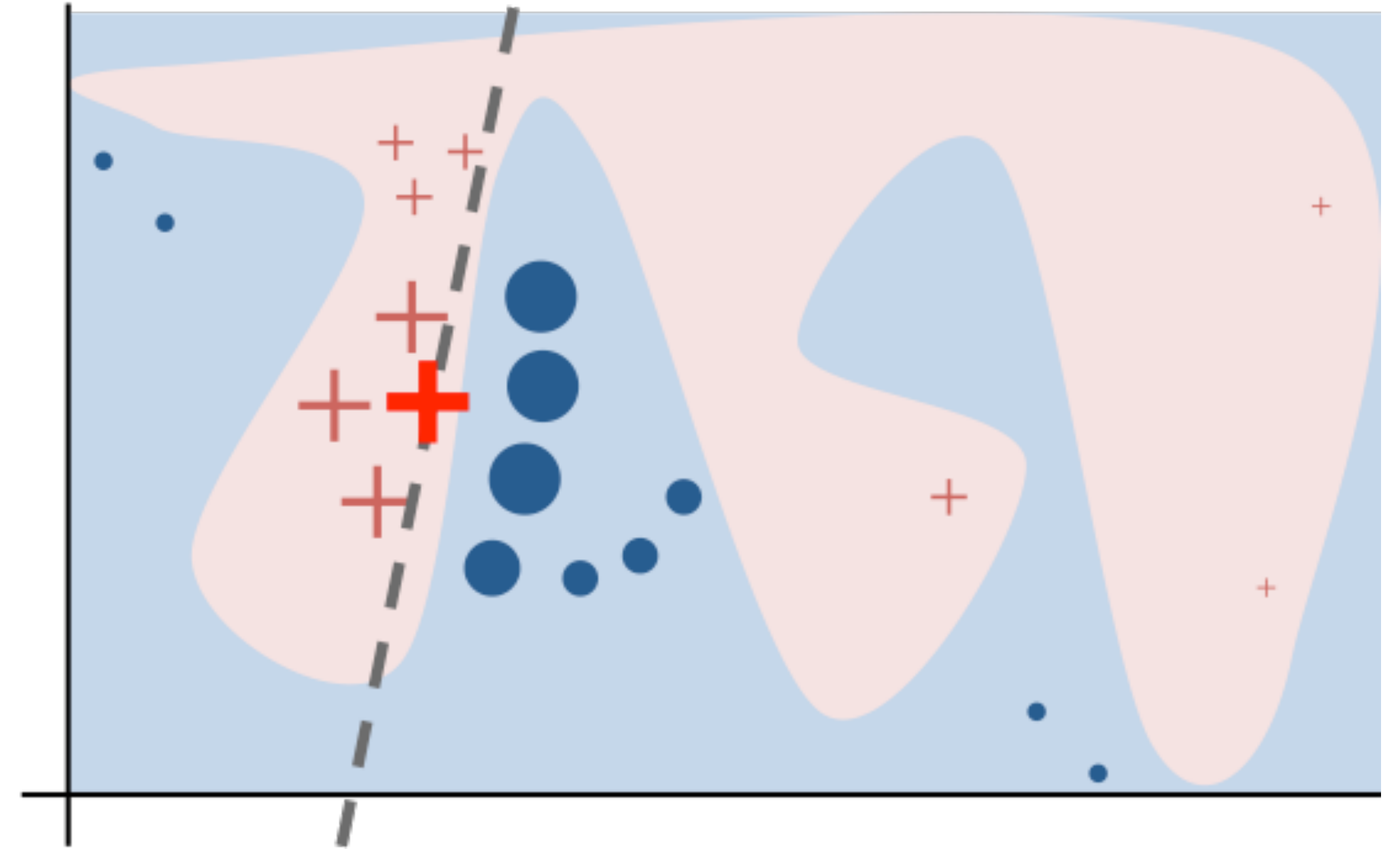


Local Interpretable Model-agnostic Explanations

For LIME, "**Explanation**" is a **local linear** approximation of the model's behaviour. While the model may be very complex globally, it is easier to approximate it around the vicinity of a particular instance.

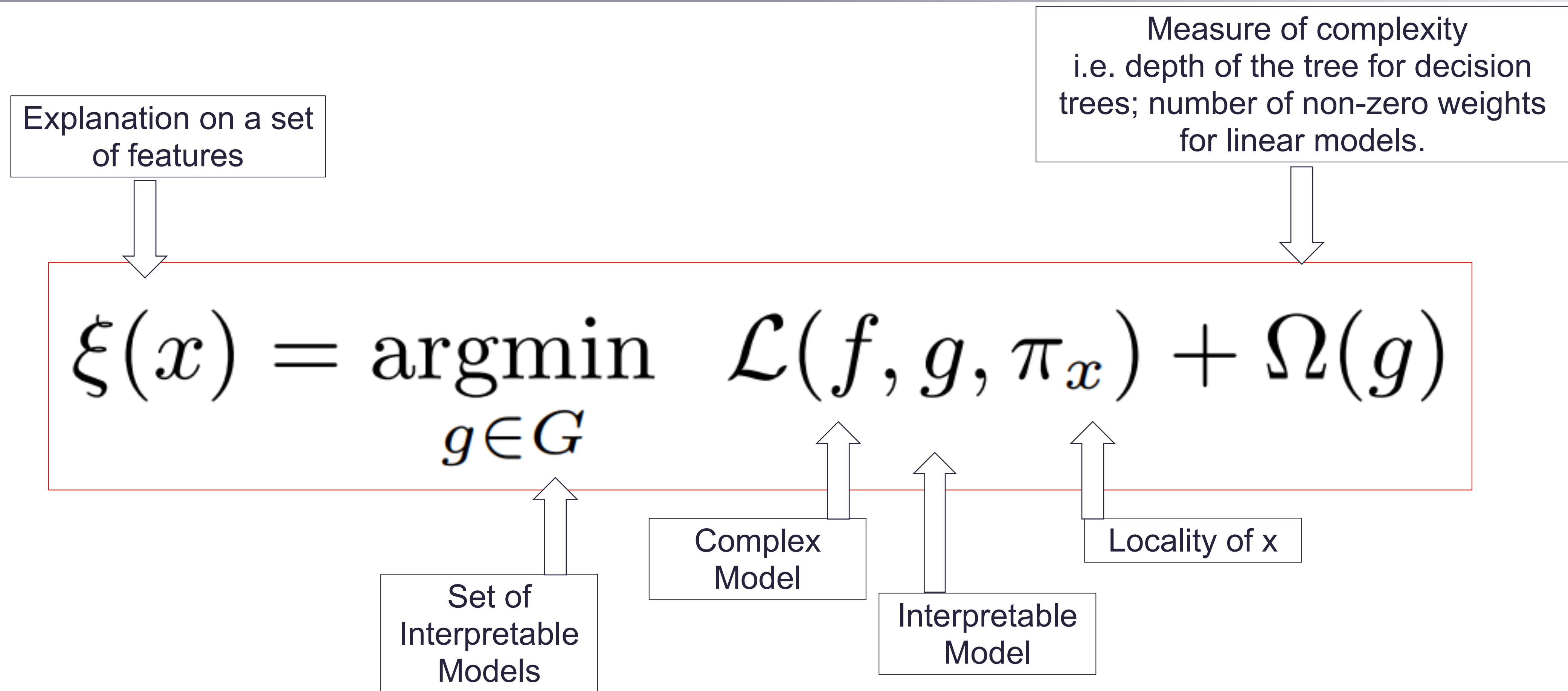
While treating the model as a **black box**, LIME perturbs the instance we want to explain (X) and learn a sparse linear model around it, as an explanation.

LIME sample instances around X , and weight them according to their proximity to X (weight here is indicated by size). LIME then learn a linear model (dashed line) that approximates the model well in the vicinity of X , but not necessarily globally.



The model's decision function is represented by the blue/pink background and is clearly nonlinear. The bright red cross is the instance being explained.

Local Interpretable Model-agnostic Explanations



Sampling for Local Exploration

To keep the explainer model-agnostic, one has to minimize the locality-aware loss function below, without making any assumptions on f . Thus, in order to learn the local behavior of f (as the interpretable inputs g vary), approximate the function by drawing samples, weighted by π of x .

$$\mathcal{L}(f, g, \pi_x) = \sum_{z, z' \in \mathcal{Z}} \pi_x(z) (f(z) - g(z'))^2$$

where:

- f is the black-box model that we are trying to explain.
- g is the local surrogate model (usually a simpler model, such as a linear regression or a decision tree).
- π_x is a proximity measure that assigns higher weights to data points z that are closer to the instance x being explained.
- $\mathcal{Z}$ is the set of perturbed samples generated around the instance x .

How does LIME work?

In short, LIME works by:

1. Perturbation of the input data (usually uniform or gaussian around
2. Prediction of perturbed samples
3. Weighting of perturbed samples based on their similarity to the original input (e.g. by an exponential smoothing kernel)
4. Training of an interpretable model (e.g., linear model) on the weighted samples
5. Interpretation of the local model

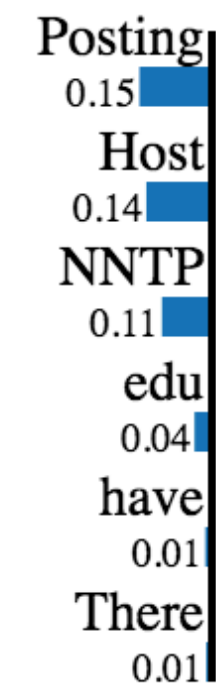
Example - Text Classification Models

Prediction probabilities



atheism

christian



Text with highlighted words

From: johnchad@triton.unm.edu (jchadwic)
Subject: Another request for Darwin Fish
Organization: University of New Mexico, Albuquerque
Lines: 11
NNTP-Posting-Host: triton.unm.edu

Hello Gang,

There have been some notes recently asking where to obtain the DARWIN fish.
This is the same question I have and I have not seen an answer on the net. If anyone has a contact please post on the net or email me.

[M.T. Ribeiro, et al. "Why Should I Trust You?" Explaining the Predictions of Any Classifier, 2016.](#)

Negative (blue) words indicate atheism, while positive (orange) words indicate christianism. The way to interpret the weights by applying them to the prediction probabilities. For example, if we remove the words Host and NNTP from the document, we expect the classifier to predict atheism with probability $0.58 - 0.14 - 0.11 = 0.31$.



Example - Image Classification Models



(a) Original Image



(b) Explaining *Electric guitar*



(c) Explaining *Acoustic guitar*

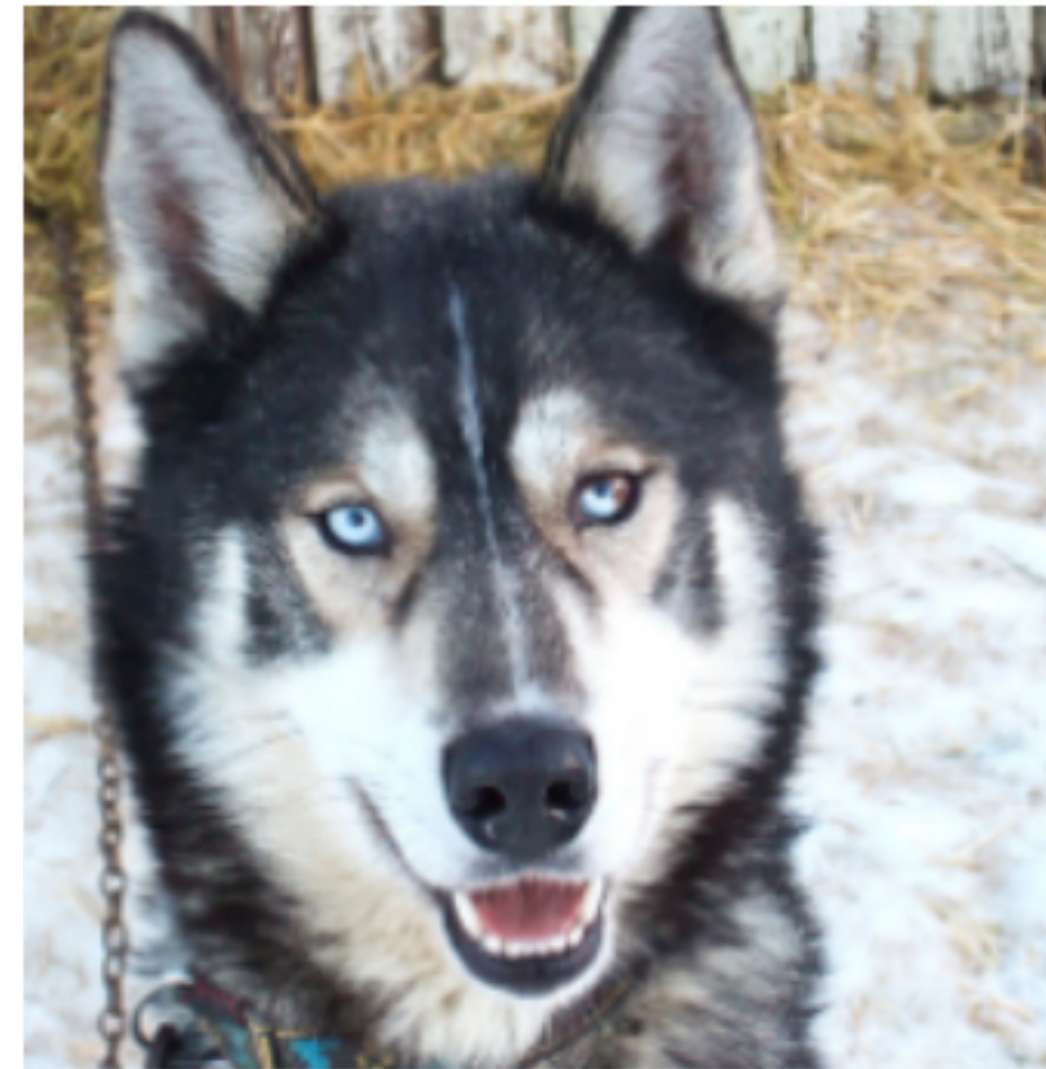


(d) Explaining *Labrador*

[M.T. Ribeiro, et al. "Why Should I Trust You?" Explaining the Predictions of Any Classifier, 2016.](#)

Explaining an image classification prediction made by Google's Inception neural network. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p = 0.24$) and "Labrador" ($p = 0.21$)

Example - Image Classification Models



(a) Husky classified as wolf



(b) Explanation

[M.T. Ribeiro, et al. "Why Should I Trust You?" Explaining the Predictions of Any Classifier, 2016.](#)

Explanation of a bad model's prediction in the "Husky vs Wolf" task. The model is considering the presence of snow in the Picture.

Benefits:

- Model-agnostic
- Provides local interpretability
- Enhances trust and transparency

Limitations:

- May not be globally faithful
- Computationally intensive
- Sensitivity to perturbation strategy

- **ANCHORS: High-Precision Model-Agnostic Explanations**

